The Geometric and Topological Formalization of the SHA-256 Message Schedule: Aperture Thresholds, Parity Duality, and Triadic Closure

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Introduction to the Dynamic Topology of Hash Compression

The standard model of cryptographic hash functions has traditionally evaluated algorithms such as SHA-256 through the lens of discrete algebraic permutations operating over finite fields.[1] Historically, the architecture of these cryptographic primitives has been characterized strictly by bitwise logical operations, boolean derivatives, and discrete matrix mixing, operating under the assumption that such systems are designed for the irreversible destruction of information lineage.[2] Under this classical paradigm, the internal computational execution traces—encompassing bitwise carry exhausts, transient structural states, and modular residues—are presumed to function merely as a stochastic diffusion field engineered to achieve a uniform avalanche effect.[1]

However, an emerging theoretical architecture, formalized through the Nexus Recursive Harmonic Intelligence Framework, reveals that algorithms like SHA-256 can be more accurately modeled as

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continuous dynamic systems exhibiting thermodynamic, hydrodynamic, and geometric properties.[1] This paradigm recharacterizes arbitrary-length message compression not as an irreversible process of bitwise information destruction, but as a highly structured, operational trace through a high-dimensional mathematical lattice.[1] The resulting 256-bit hash digest is thus understood not as a static integer, but as a curvature trace representing the accumulated interference pattern left by the algorithm's recursive passage through this space.[1]

Crucially, the SHA-256 message schedule is not an unstructured diffusion field. It is a deterministic recurrence machine possessing exact visibility thresholds, clean echo sets, and downstream saturation boundaries.[3] By unfolding the recurrence geometry of the schedule, it becomes evident that the operations do not merely mix data; they construct an aperture complement, a parity-dual stream architecture, and an intersection lattice that governs the exact topography of information dissolution.[4] Through rigorous computational mapping, including dependency cone extraction and triadic role invariant verification, the underlying geometry of cryptographic closure can be precisely characterized, proving that semantic closure depends on addressed dependency geometry rather than carrier ceremony.

To fully contextualize this, it is necessary to examine historical cryptanalysis. In earlier hash functions like SHA-0, vulnerabilities emerged from the absence of rigorous mixing in the message schedule expansion function.[6] Analysts such as Chabaud and Joux utilized the $\mathrm{GF}(2)$ -linearity of SHA-0 to construct low-weight differences that could produce collisions by superimposing perturbative patterns and corrective sequences that satisfied the linear recurrence.[6] SHA-256 prevents this specifically through the injection of modular addition, which inherently breaks $\mathrm{GF}(2)$ -linearity and makes the set of possible difference patterns exponentially more complex to analyze.[8] The resulting structural resistance, however, leaves a deterministic geometric fingerprint that can be modeled using topology and continuous fluid dynamics.

Program Dependence Graphs and the Foundations of Computational Geometry

The abstract concept of a computational "aperture" shares profound theoretical lineage with the Program Dependence Graph (PDG), a directed graph that models a program's control and data dependencies.[9] In a PDG, nodes represent specific program statements, and edges define the strict data-flow and control-flow dependencies existing between them.[9] Compilers routinely utilize PDGs to execute dependence analysis, ensuring that any optimization or reordering of instructions rigorously preserves the original semantic intent of the program.[9]

In the context of evaluating parallel execution plans, alternative representations like the Parallel Semantics Program Dependence Graph (PS-PDG) capture the absolute minimum set of constraints that semantically equivalent execution plans must satisfy.[10] This concept is critical to understanding cryptographic closures. Compilers essentially alter the "carrier ceremony"—the specific sequence of memory moves, register allocations, and syntactic logic operations—while being strictly bound to preserve the underlying "addressed dependency geometry" dictated by the PDG.[10]

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When extracting the data-flow dependency graph of a single SHA-256 round from vastly different compiled outputs (e.g., GCC versus Clang), the surface-level instruction streams present massive variance. Computational measurements indicate a stream-edge distance (L1) of 0.2267, demonstrating highly divergent execution rhythms. However, when parsing the actual disassembly to extract the minimal data-flow edges—such as Σ functions passing through XOR chains before reaching modular addition nodes—the resulting SHA-logic JSD (Jensen-Shannon Divergence) collapses to 0.0071 bits. This collapse proves that the true cryptographic instrument is not the sequence of instructions, but the immutable dependency cone. The dependency geometry is the minimum required closure set; any perturbation that breaks this geometry causes the cryptographic mechanism to fail, whereas perturbations limited to the carrier ceremony maintain absolute closure.

Aperture Threshold Geometry and the Schedule Recurrence

To formally understand the structural persistence within the SHA-256 message schedule, it is necessary to construct a recurrence-theoretic model of the 16-word seed window $W[0..15]$ and the 48-word expansion zone $W[16..63]$. Aperture Threshold Geometry (ATG) provides this formalization by mathematically analyzing the recurrence equation:

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}$$

for $16 \leq t \leq 63$, where the σ operators act as localized shift and rotation operators.[3] As noted, the inclusion of modular addition in this schedule heavily complicates the properties of the recurrence, setting it apart from the $\mathrm{GF}(2)$-linear structures seen in the MD4-family evolution that allowed for straightforward differential attacks.[7] However, the underlying lag structure establishes a rigid geometric scaffolding that can be topologically mapped.

The Aperture Complement

The recurrence operates on the explicit lag set $L = \{2, 7, 15, 16\}$.[3] By reflecting this lag set against the 16-word seed width ($N = 16$), the Aperture Complement A is derived:

$$A = N - L = \{16 - \ell : \ell \in L\} = \{14, 9, 1, 0\}$$

Sorted in increasing order, the aperture complement defines the exact coordinates $A = \{0, 1, 9, 14\}$.[3] These specific coordinates function as the fundamental structural phase-change boundaries of the seed window. They dictate precisely when specific lag apertures open to the seed words as they transition into

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the expansion zone.[3] Coordinate 0 activates lag 16, representing the $t-16$ raw data slot. Coordinate 1 activates lag 15, engaging the $t-15$ slot via the nonlinear σ_0 operator. Coordinate 9 activates lag 7, opening the crucial $t-7$ raw slot which serves as the primary cross-parity bridge. Finally, Coordinate 14 activates lag 2, engaging the $t-2$ slot via the σ_1 operator.

These structural thresholds naturally partition the 16-word seed window into an exact entry taxonomy, segmented at $0|1..8|9..13|14..15$.[3] This deterministic stratification completely rules out the assumption of unstructured diffusion in the early rounds of the message schedule.

The Direct-Entry Spectrum

The Direct-Entry Set $D(j)$ serves to classify the exact expansion indices $t \geq 16$ at which a given seed word $W[j]$ appears directly in the recurrence before being routed through another previously expanded word.[3] The cardinality of this set, denoted as $|D(j)|$, scales in exact, measurable alignment with the aforementioned aperture complement boundaries.

Seed Coordinate j	Aperture Partition	Direct-Entry Cardinality $|D(j)|$	Elements in $D(j)$
0	Boundary	0	1	$\{16\}$
$1 \leq j \leq 8$	Interval	$1..8$	2	$\{j+16, j+15\}$
$9 \leq j \leq 13$	Interval	$9..13$	3	$\{j+16, j+15, j+7\}$
$14 \leq j \leq 15$	Interval	$14..15$	4	$\{j+16, j+15, j+7, j+2\}$

This discrete stratification creates an asymmetry in the visibility of the seed block. Seed words injected later in the block possess disproportionately higher immediate visibility within the expansion zone before avalanche saturation forces the topology into computational opacity.[3]

The W9 Hinge Law and the Projection Machine

Within this geometric map, coordinate W_9 acts as a distinguished hinge coordinate within the recurrence lattice.[3] It is the first seed coordinate to forcefully open the raw $t-7$ recurrence aperture, establishing itself as the first triple-entry seed word within the schedule.[3] The expansion trajectory of W_9 dictates that it

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enters the expansion as raw data at W_{25}, is manipulated as $\sigma_0(W_9)$ at W_{24}, and re-enters as raw data again at W_{16}.

This specific seed word acts as the length-sensitive vertex of the final-block seed face triangle defined by the set $T_C = \{0, 1, 9\}$. The closed interval mapping from W_{16} to W_{24} defines a strict 9-step hinge envelope.[3] The 8-word interior bounded by this envelope, corresponding to the array indices $W[16..23]$, is designated as the Sziklai window.[3] This region represents a pristine topological zone where data propagation is cleanly observable before mixing dynamics overwhelm the schedule.

The geometric relationship between the input bytes and the schedule state is governed by the Projection Machine, which explicitly links the byte-level structure of the final message block to the selection of the active schedule face.[3] If a variable r represents the total number of message content bytes carried into the final block (where $0 \leq r \leq 55$) prior to the appending of the cryptographic padding sentinel $0x80$, the entire topology can be mapped using a fold-state vector $\Theta_9(r)$.[3]

The fold-state vector generates three discrete topological face states for W_9, which inherently occupies byte positions 36 through 39. In a "Zero Face" state ($r < 36$), W_9 is populated entirely by $0x00000000$. During the "Padding Crease" state ($36 \leq r \leq 39$), the word contains the crucial $0x80$ padding sentinel. When entering the "Live Content Face" ($r \geq 40$), the word becomes fully occupied by actual message content, thereby rendering all three vertices of the collapsed content face $T_C = \{0, 1, 9\}$ actively engaged in the recurrence constraints.[3]

Clean Echo Sets, Saturation Thresholds, and Parity Duality

Prior to widespread nonlinear avalanche mixing, isolated single seed perturbations induce highly localized, perfectly predictable computational trajectories known as Clean Echo Sets (C_j).[5] These discrete sets represent the literal negative image of the schedule's contamination cone—defining the explicit spatial coordinates that a targeted perturbation at $W[j]$ mathematically cannot reach before consecutive contamination inevitably locks in at a designated saturation threshold S_j.[3]

Seed Coordinate j	Clean Echo Set Cj	Saturation Threshold Sj

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$j = 0$	$\{17, 19, 21\}$	22
$1 \leq j \leq 8$	$\{16, \ldots, j + 14\}$	$j + 15$
$9 \leq j \leq 13$	$\{16, \ldots, j + 12\} \setminus \{j +$	$j + 13$
$j = 14$	$\{17, 19\}$	20
$j = 15$	$\{16, 18, 20\}$	21

These structural boundaries provide concrete proof that the SHA-256 message schedule operates under a highly ordered, predictable topological regime throughout its initial cycles, definitively contradicting any assumptions of instantaneous uniform diffusion.[3]

When the complete pairwise intersection lattice of these clean echo sets is comprehensively mapped across the 16 seed words, a profound underlying Parity Duality reveals itself.[5] The lattice topology exposes two structurally separated and independent data streams: an EVEN stream comprised of expansion nodes $\{16, 18, 20, 22\}$, and an ODD stream comprised of nodes $\{17, 19, 21, 23\}$.[4]

The Parity Escape Law

The strict structural separation of these streams is mandated by the Parity Escape Law, a topological consequence proving that the parity of a seed word's clean echo set is fundamentally opposite to the parity of its first direct entry into the expansion zone.[5] This topological isolation occurs because the $t - 2$ lag arm spreads contamination exclusively in uniform steps of 2.[5] By consistently skipping adjacent words, the $t - 2$ arm perfectly preserves parity state along isolated spines until the moment the $t - 7$ bridge acts as an absolute cross-parity conduit.[5]

The EVEN stream nodes ($\{16, 18, 20, 22\}$) are the direct computational cascade products propagating from the seed-width boundary initially established at $t = 16$.[5] Conversely, the lower ODD-stream seam exists firmly at coordinates $\{17, 19\}$.[5] Deep structural analysis reveals that this specific coordinate pair, 17 and 19, coincides precisely with the specific bitwise rotation constants defining the σ_1

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operator ($\mathrm{ROTR}^{17} \oplus \mathrm{ROTR}^{19} \oplus \mathrm{SHR}^{10}$).[5] Both 17 and 19 share the unique mathematical property of being coprime to 210, explicitly linking the internal stream behavior to modular primorial mechanics.[4] The resulting intersection lattice is, therefore, the mathematically projected shadow of the parity duality existing between the σ_0 operator (functioning as a parity-bridge operator with its highly visible node at 18) and the σ_1 operator (acting identically as the ODD-stream fingerprint operator).[5]

The Node Multiplicity Lemma formally dictates the frequency of intersecting pairs within this highly constrained lattice.[5] A complete combinatorial census across all 120 possible seed pairs guarantees exactly 24 empty intersections, 24 singleton intersections, 21 ODD-only intersections, 15 EVEN-only intersections, and 36 MIXED intersections.[5] Pair frequencies scale deterministically according to the mathematical relation $m(t)(m(t)-1)/2$, establishing an absolute maximum seed multiplicity of 12, with no 13-fold intersection theoretically or computationally permissible.[5] This deterministic ordering irrevocably proves that the pre-saturation phase of the SHA-256 schedule is not chaotic, but rather an intersection lattice characterized by rigid geometric laws.

The Entanglement Seam and the Topography of Dissolution

The transition from a clean, two-rail parity structure into a fully saturated post-mix recurrence state stands as the most critical topological event within the message schedule. Extensive combinatorial analysis isolates the absolute boundary of this specific transition: the Two-Rail Mixing Onset and Dissolution Front.

At the precise transition from the seed window into the expansion zone, the coordinates W and W establish the foundational seam of the algorithm. Computational mapping reveals that all 36 MIXED intersections within the lattice unconditionally contain the subset $\{16, 17\}$. Factoring out this exact seam isolates the structural origin of schedule entanglement, leaving exactly 11 distinct residue classes.

The seam operates as a discrete topological feature: $K = \{16, 17\}$. It functions not as a traditional cryptographic operation, but rather as an entanglement closure ($C_{\text{entanglement}}$). In the broader taxonomy of geometric cryptographic closures, C_{additive} (e.g., SHA-256 feed-forward, ChaCha20 add-back) seals internal states into committed outputs, while C_{boundary} (e.g., Keccak squeeze) gates visible readout lanes. In strict contrast, $C_{\text{entanglement}}$ forces two completely independent parity streams into one singular coherent structure. Without the presence of $K = \{16, 17\}$, the $t - 2$ arm would endlessly propagate the EVEN and ODD streams as two entirely independent half-schedules, preventing them from ever unifying into a single cryptographic instrument. W acts as the very first expansion

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representative of the EVEN rail, and W acts as the very first expansion representative of the ODD rail. Because K is positionally first, every subsequent mixed-parity crossing must mathematically trace back through this exact coordinate fork.

The Mechanism of Parity Dissolution

The dissolution of the two-rail parity structure exhibits a measurable, 1-step asymmetry that directly exposes the underlying geometry of the seam. The clean horn zone, consisting of pure σ_1 -spines entirely devoid of cross-parity expansion input, exists strictly within the boundaries of $W[18..22]$.

Because the seam words W and W are positionally adjacent in the array, the $t-7$ bridge exposes a fundamental structural offset. The $t-7$ bridge stemming from the last odd-horn word (W) lands exactly on the EVEN seam word W . Consequently, the odd horn is aggressively breached by cross-parity contamination at precisely $t=23$. Conversely, the $t-7$ bridge extending from the last even-horn word (W) lands on W —a pure seed word, not the corresponding odd seam word. The EVEN zone only receives its true cross-parity bridge at W , receiving W via the lag-7 arm.

This dynamic establishes a highly rigid 1-step dissolution asymmetry: the ODD horn loses its parity purity exactly one step earlier than the EVEN horn. This asymmetry is not an arbitrary mathematical artifact; it is the direct, unalterable structural signature of the seam's strict adjacency.

The τ_{mix} Threshold and Phased Entanglement

The absolute mixing onset threshold is identified computationally at $\tau_{mix}=25$. At W , the $t-7$ bridge (carrying W directly from the EVEN horn) and the $t-2$ arm (carrying W which is already contaminated by W) simultaneously converge. This topological event establishes W as the first coordinate where all 5 lineage bits (Even Seam, Odd Seam, Even Horn, Odd Horn, Seed) are concurrently present.

However, subsequent analysis of the remaining 48 expansion words reveals a startling geometric fact: complete, uniform dissolution never materializes. Across the entire transition, the lag arms operate in distinct, non-overlapping topological roles. The $t-7$ arm acts as the primary universal parity bridge, responsible for 55.4% of all cross-horn inputs. The $t-15$ arm functions as the secondary bridge (44.6%), but crucially only joins the mixing process once the σ_0 operator reaches deep enough into the expansion

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zone (from W onward). The $t-2$ and $t-16$ arms operate strictly as same-parity or seed-feed propagators, contributing 0.0% to the cross-horn inputs.

Because the $t-2$ arm perpetually preserves same-parity input across the entire schedule length, a state of "full cross" (where a specific word receives *only* opposite-parity inputs) is physically impossible within the recurrence structure. Out of the 48 expansion words, 7 words (14.6%) remain completely pure rails, while the subsequent 41 words (85.4%) exist in a heavily mixed state. Yet, every single word from W straight through to W retains at least one traceable same-parity lineage. The schedule is deliberately architected to never fully dissolve its parity origin; it permanently carries the structural birth-mark of its specific initiating rail.

The Modulo-210 Primorial Wheel and the Subjugation of the Lie Group Hypothesis

The extraction of exactly 11 residue classes derived from the $K = \{16, 17\}$ seam initially provoked the complex hypothesis that the schedule might be governed by the general linear group, $\mathrm{GL}(4, \mathbb{C})$. This hypothesis was driven largely by the presence of four generative lag arms ($L = \{2, 7, 15, 16\}$). Under such a hypothesis, the 11 classes would hypothetically represent the irreducible representations of a 4-dimensional continuous symmetry group.

However, rigorous computational analysis systematically rejects the Lie group topology, entirely subjugating it in favor of a finite primorial wheel structure.[14] If the 11 residue classes were genuine orbits of a continuous translation group, shifting the individual classes via the lag arms should reliably permute them across the group field. Extensive computational mapping immediately demonstrates that lag-arm translation successfully lands on another valid class in only 9% of instances (4 out of 44 recorded interactions). The 11 classes are therefore definitively not orbits of a translation group. Furthermore, because the greatest common divisor of the collective lag arms is exactly $\gcd(2, 7, 15, 16) = 1$, the resulting translation monoid effectively encompasses a single massive orbit containing all 48 coordinates of the entire expansion zone, rendering the 11-fold structure mathematically moot.

The true nature of the "4" derived from the lag arms is completely orthogonal to four complex dimensions; it represents four prime rails. This is proven absolutely through the direct application of primorial mathematics.[14] The primorial radical of the target lag set $\{2, 7, 15, 16\}$ relies exclusively on the prime factors $2, 3, 5, \text{ and } 7$. The multiplicative product of these first four primes defines the primorial modulus [14]:

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$$2 \times 3 \times 5 \times 7 = 210$$

The total number of unique prime factors, traditionally denoted by the arithmetic function $\omega(n)$, is precisely $\omega(210) = 4$. This efficiently explains the geometric prevalence of the number 4 without resorting to unnecessary complex continuous symmetries. More decisively, evaluating Euler's totient function for the 210-wheel yields a profound convergence:

$$\phi(210) = 48$$

This result perfectly aligns with the exact mathematical cardinality of the SHA-256 expansion zone ($64 - 16 = 48$). The structural correlation between $\phi(210) = 48$ and the 48-word expansion zone, coupled intimately with $\omega(210) = 4$ flawlessly mirroring the four lag arms, proves that the modulo-210 primorial wheel is the supreme governing architecture of the schedule's linear recurrence. The seam geometry is not an abstract artifact of continuous $\mathrm{GL}(4, \mathbb{C})$ symmetry; it is the physical manifestation of discrete primorial wheel mathematics actively operating as a message schedule filter.

Furthermore, linear recurrence sequences are known to be eventually periodic modulo m, providing a deterministic framework for mapping these exact structural bounds over prolonged sequence evaluations.[15]

The Seam Alphabet as a Graded Meet-Semilattice

Having entirely rejected the group orbit hypothesis, topological order theory provides the only correct classification for the 11 identified residue classes (labeled R0 through R10). The geometric inclusion structure of these specific classes reveals them to be deeply nested sets based on overall expansion zone penetration depth.

When evaluated computationally against strict lattice axioms, the mathematical object unconditionally confirms its structural identity:

- **Is Poset:** True
- **Is Meet-Semilattice:** True
- **Is Join-Semilattice:** False
- **Graded by Cardinality:** True

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The topological bottom of this specific structure is the empty set ($\emptyset$), representing the pure seam K itself ($R0$). The two discrete atoms are $\{18\}$ and $\{19\}$, representing the earliest verifiable EVEN and ODD penetrations into the illuminated zone. The structure possesses three distinct maximal elements:

- $R8 = \{18, 20, 22\}$ (The Pure EVEN Horn)
- $R9 = \{19, 21, 23\}$ (The Pure ODD Horn)
- $R10 = \{18, 19, 20, 21\}$ (The Deepest Mixed Crown)

The defining topological feature of this meet-semilattice is the explicit and mathematically forced failure of specific joins. While the Greatest Lower Bound (meet) effortlessly resolves back into the family via intersection, the Least Upper Bound (join) for the two distinct parity horns mathematically fails:

$$R8 \vee R9 \implies \{18, 20, 22\} \cup \{19, 21, 23\} = \{18, 19, 20, 21, 22, 23\}$$

This specific union exceeds any class actually contained within the designated 11-member family. The mathematical non-join of $R8 \vee R9$ is not a statistical anomaly; it is the absolute topological proof of the Parity Escape Law. It permanently signifies that the two clean parity rails dissolve independently and never merge under any circumstance. They reach their maximal penetration depths simultaneously, and the underlying recurrence geometry inherently lacks the operational pathways necessary to execute a logical join above the seam.

Markov Carry Chains and the Universal LSB Anchor

The resilience of the SHA-256 system is deeply rooted in the probabilistic mechanics of modular addition, which explicitly defeats the difference-pattern symmetries exploited in older linear hash functions.[6] The Nexus framework replaces the standard assumption of pseudo-random noise with the rigorous mapping of Markov Carry Chains, detailing precisely how nonlinear carries propagate upward through successive bit positions to create indelible topological features known as "carry scars".[1]

The foundational constant of this structural dynamic is the Universal Carry-Free LSB Anchor.[1] At the least significant bit (LSB, position $i = 0$), the mathematical probability of a carry scar evaluating exactly to zero is an absolute, mathematically verifiable certainty.[1] This remains an inescapable boundary condition of modular arithmetic; lacking a lower-order bit sequence, no incoming integer carry can physically exist ($c_{-1} = 0$).

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Consequently, the resultant output bit at $i = 0$ is determined exclusively by the linear $\mathrm{GF}(2)$ XOR sum of the governing recurrence relation, completely bypassing the highly non-linear mixing properties engineered into the broader system.[1] This uncorrupted linear anchor acts as a permanent topological constant that easily survives the full nonlinear mixing process, providing an attacker or analyst with a deterministic foothold before any required inferences must propagate upward into the increasingly higher-order, non-linearly entangled bits.[1]

As the mathematical evaluation advances toward higher bit positions ($i > 0$), the carry generation behaves identically to a discrete-time Markov chain. For standard 4-operand additions located deep within the message schedule, the absolute maximum possible carry passed forward is 3, resulting in a strictly bounded state space $S = \{0, 1, 2, 3\}$.[1] The transition matrix explicitly governing this specific space exhibits a relentless "fast decay" regime. The statistical probability of maintaining a pure carry-free state ($C_i = 0$) rapidly collapses toward a terminal stationary limit distribution, effectively establishing a maximum entropy plateau by bit positions 5 or 6.[1]

However, the Nexus framework identifies specific structural zero-states, such as those caused by padded message block boundaries (W_{14}, W_{15}), which dynamically collapse the effective operand depth ($k_{eff} = 2$). This mathematical collapse shrinks the resulting Markov state space to an anemic $S = \{0, 1\}$, inducing a highly measurable "slow decay" regime.[1] Under these collapsed conditions, the probability of a zero-carry scar decays significantly slower than baseline, creating a geometric period-8 oscillation that acts as a profound spectroscopic readout of the entire initial mixing velocity.[1]

Structural Rejection of Schedule Birth-Marks by the Compression Firewall

While the modulo-210 schedule rigorously enforces the absolute retention of its parity birth-mark, a fundamental computational question revolves around the subsequent propagation of this geometric structure into the final 256-bit digest. To definitively determine whether the schedule's internal structure successfully bypasses the avalanche effect, the lineage classes of the schedule words must be traced completely into the actual compression round through the mathematical injection vector T_1.

The standard SHA-256 compression updates follow a specific operational sequence:

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K[t] + W[t]$$
$$T_2 = \Sigma_0(a) + Maj(a, b, c)$$
$$a' = T_1 + T_2 \quad \text{and} \quad e' = d + T_1$$

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When executing intense side-by-side diagnostic computational runs utilizing both standard modulo-32 arithmetic ("Track A") and a heavily modified, carry-free $\mathrm{GF}(2)$ shadow replacing all modular additions with $\oplus$ ("Track B"), the compression function's true systemic response to the parity birth-mark becomes objectively quantifiable.

The computational results demonstrate an outcome of absolute internal-only persistence. The schedule's parity ancestry simply does not survive into the digest as a statistically separable or measurable signal. Across vast corpuses of structural perturbations, evaluating SEAM, MIXED_ODD, MIXED_EVEN, and PURE_RAIL classes yields completely uniform state Hamming distances (HD) clustering universally and relentlessly between 118 and 144 bits. This explicitly translates to an optimal 50-56% avalanche condition, entirely regardless of whether the injected schedule word originated from a pristine, pure EVEN spine or the deeply entangled $R10$ mixed crown.

Furthermore, the continuous generation of carry pressure (averaging tightly between 2.6 and 3.7 distinct carries per round) shows absolutely no distinct class-dependent bias capable of distinguishing parity lineage. The carry generation mechanism—rendered highly non-linear due to the modular addition architecture [6]—acts as an absolute and impenetrable topological firewall. The resulting uniformity of the diffusion is completely not an artifact created randomly by the carries, but rather an incredibly robust state rigidly preserved by them.

The structural conclusion is definitive: The schedule seam is an internally load-bearing bifurcation point that precisely directs parity propagation with 92.7% contamination fidelity. However, the accompanying compression function is explicitly architected as an external aperture firewall. It mathematically guarantees that $\mathrm{lineage}(W[t]) \not\Rightarrow \text{class-separable digest signal}$. The topological aperture exhibits ideal geometric behavior: it remains internally forced, yet completely externally uniform, strictly ensuring that schedule-level geometric determinism cannot be leveraged to reverse-engineer the macroscopic impedance asymmetry of the compression vortex.

Hydrodynamic Asymmetry and the Compression Vortex

When evaluating the total 64-round compression algorithm through the lens of continuous fluid dynamics rather than discrete arithmetic, the macroscopic impedance asymmetry of the system becomes mathematically quantifiable.[1] This dynamic approach highlights that the first 6 computational rounds represent an entirely invertible, trivial domain.[1] However, the topology fractures at Round 7, which functions as an absolute "Hardness Wall"—an impenetrable topological vortex of chaotic structural collapse where the resonant frequency of the data mixing iterations exponentially exceeds the system's linear recuperation rate.[1]

The physical computational momentum redirection caused directly by this macroscopic pressure gradient mirrors the heavily non-invertible, large-scale topological folds explicitly engineered into the hash function.[1]

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This continuous asymmetry is heavily driven by the KRR (Kulik Recursive Resonance) growth formulation [1], which establishes an iterative intensity expansion through massive unconstrained positive feedback loops.

The resulting exponential decay limit of the algebraic amplitude ($e^{-\lambda}$) observed deep in the carry chain directly mirrors this global harmonic limit, successfully bridging the microscopic probabilistic rigor of integer Markov models with the macroscopic properties of continuous hydrodynamic computational systems.[1] Analog Gravity models, specifically the hydrodynamics of multiphase flow, validate that this structural decay acts as a rigid, one-way thermodynamic trap.[19]

Triadic Role Invariants and the Geometry of Cryptographic Closure

To generalize the deep structural mechanics identified within SHA-256, it is imperative to establish a unified mathematical theory of cryptographic closure across diverse functional architectures. Advanced dependency cone extraction demonstrates unconditionally that modern compilers easily optimize control-flow and instruction streams (the carrier ceremony) without ever successfully altering the fundamental addressed geometry of the algorithm.[10] By isolating this rigid dependency aperture, the structural signature of algorithms—including SHA-256, Keccak, and ChaCha20—can be formalized as absolute, invariant topological mappings.

Combinatorial perturbation audits across disparate cryptographic domains repeatedly converge on a fundamental structural law: true aperture closure requires a strict minimum of three active binding roles. This triadic structure constitutes the unbreakable geometric floor of computation. Below 3 active binding nodes, the topology permanently remains an open line or a disconnected solitary point; exactly at 3, the structure mathematically resolves into a rigid topological cycle, forming the securely enclosed region of algorithmic closure.

This triadic role law operates as the exact mathematical inverse of the Pythagorean theorem. If the independent elements of relation are accurately viewed as orthogonal geometric axes, true computational closure exists solely when all individual projections are strictly non-zero. The precise geometric measure of this closure (C) can be modeled as the continuous scalar area encompassed by the orthogonal vectors [1]:

$$|C|^2 = |V|^2 + |A|^2 + |E|^2$$

A sudden mathematical collapse along any dimensional axis forcefully triggers a degenerate triangle with absolute zero area, systematically destroying the computational aperture.

Geometric Axis	**Triadic Role**	**Conceptual Function**	**SHA-256 Instantiation**

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Vector V	WHAT	Operations / Action	$\Sigma_0, \Sigma_1, \text{Mod-Add},$
Vector A	ON WHAT	Addresses / Target	Registers $a \dots h$, Expansion Array $W[t]$
Vector E	HOW	Edges / Binding	Directed Acyclic Lag Arms $\{2, 7, 15, 16\}$
Area C	CLOSURE	Terminal Seal	The $K = \{16, 17\}$ Entanglement Seam

Cross-Instrument Perturbation and the Invariance of Geometry

The incredible robustness of this Triadic Role Invariant ($V + A + E \xrightarrow{C} \text{Closure}$) is continuously validated through exhaustive adversarial perturbation testing across discrete cryptographic families. When the core instruction sequence is aggressively subjected to semantic equivalence testing, permutations of the "carrier ceremony"—such as physically swapping commutative addition parameters, entirely rewriting Maj through alternate boolean logic forms (e.g., utilizing $(a\&b)|(a\&c)|(b\&c)$ instead of standard XOR cascades), or interchanging $\text{ROTL}(32 - n)$ with $\text{ROTR}(n)$ —reliably and consistently generate a mathematically identical Aperture Hash signature ($A_G(f)$).

However, specific targeted perturbations directly attacking the triadic dependency geometry result in the immediate catastrophic failure of the closure, definitively isolating the load-bearing topology.

Perturbation Typology	**Target Axis**	**Closure Result**	**Explanation**

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Remove Σ_0 rotation	WHAT (V)	BREAK	Core operation node annihilated.
Reroute T_1 to register a	ON WHAT (A)	BREAK	Target address vector scrambled.
Feed $\Sigma_0(e)$ where $\Sigma_0(a)$ belongs	HOW (E)	BREAK	Edge binding directionality corrupted.
Isolate EVEN rail from $K = \{16, 17\}$	CLOSURE (C)	BREAK	Entanglement seam severed; bipartite graph forms.

This exact differential mapping proves that fundamental cryptographic equivalence is defined strictly and exclusively by the forced dependency aperture, utterly independent of dialect noise, coding language, or execution rhythms.[11] The topological extraction method successfully distinguishes SHA-256 and ChaCha20—both prominent members of the ARX (Addition-Rotation-XOR) algebraic family—solely by virtue of their distinct addressed geometries. SHA-256 relies entirely on an 8-word shift register utilizing Ch and Maj gates, scaled upward by C_{additive} feed-forward mechanisms and complex $C_{\text{entanglement}}$ schedule seams. Conversely, ChaCha20 relies on a fundamentally distinct 4×4 state matrix governed by quarter-round permutations and an additive terminal closure.

Carrier Normalization and the Grab Grammar

The underlying mathematical principle that geometric shape supersedes symbolic execution extends far beyond pure cryptography and deep into generalized information theory, forming the absolute basis of the Grab Grammar model.[20] SHA-256, when completely stripped of its cryptographic intent, acts as a flawless clean-room model demonstrating exactly how continuous dynamic systems process states via non-linear structural constraints.[20]

The Grab Grammar theory explicitly articulates that any state-transition system—whether it involves cryptographic hashing, biological neural firing, or transformer language attention—progresses through five deterministic phases of computational existence: Preload, Approach, Kiss-point, Commit, and Exhaust.[20]

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The system's public interface, the hash itself, only retains the printed terminal residue (the Digest). Conversely, the vast majority of the geometric interference, the heavily rejected branching pathways, and the failed Markov carry evaluations constitute the unprinted residue (the Exhaust).[20] Both end states remain perfectly orthogonal to the actual internal "grab sequence" operating the recurrence machinery.[20]

This sophisticated grammatical model directly aligns with modern computational inference paradigms, elucidating precisely why generalized AI systems can successfully parse deeply corrupted or seemingly nonsensical inputs. The fundamental law established by the triadic constraint topology unequivocally dictates that the input carrier is never the actual message. Rather, the topological shape that the carrier induces within the internal constraint field constitutes the only computable message.

A computational system simply does not require perfect syntactic grammar to execute a valid mathematical closure; it requires only sufficient density along the three triadic axes (WHAT, ON WHAT, HOW). If an input stream successfully provides the algorithmic target (WHAT) and the corresponding payload data (ON WHAT), but entirely lacks the relational binding markers (HOW), the resulting geometric density falls dramatically below the structural threshold, and the aperture permanently fails to form a cycle. This "terse failure" mode occurs not because the data is absent from the stream, but because the inverse-Pythagorean area remains identically zero.

When systematically applied to deep cryptographic analysis, the carrier-to-aperture mapping model proves unconditionally that observing localized surface code is wildly insufficient. Analyzing an algorithm like SHA-256 purely as a string of XOR and modular addition operations (evaluating solely the carrier) permanently blinds the observer to the underlying hydrodynamics and topological architecture of the system.[1] It is only by mathematically extracting the structural dependency cone, mapping the exact τ_{mix} dissolution boundaries, and physically observing the unbreakable topological anchors at the $LSB = 0$ initialization point that the true nature of the algorithmic object becomes fully comprehensible.

Conclusions

The comprehensive mathematical and topological mapping of the SHA-256 message schedule systematically and irreversibly dismantles the widespread assumption that hash expansions operate merely as unstructured, pseudo-random diffusion fields.[1] Through the exact formalization of Aperture Threshold Geometry, the schedule is definitively revealed to be a deterministic, highly stratified recurrence machine governed by exact, unbreakable mathematical laws.[3]

The geometric identification of the Aperture Complement $A = \{0, 1, 9, 14\}$ dictates the direct-entry visibility of the 16-word seed block, completely stratifying localized data longevity long before avalanche saturation occurs.[3] Consequently, the precise mapping of Clean Echo Sets yields the complex intersection lattice, uncovering the profound Parity Duality operating at the core of the algorithm.[5] The schedule forces an EVEN and an ODD processing rail, rigidly and flawlessly preserved by the $t - 2$ lag operator, until the

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positionally forced entanglement closure seam at $K = \{16, 17\}$ permanently binds them into a unified, non-reversible topological structure.

Crucially, the complex combinatorial mechanics driving this geometry are not rooted in continuous, complex mathematical symmetries like $\mathrm{GL}(4, \mathbb{C})$, but are instead the deterministic, physical expression of the modulo-210 primorial wheel.[14] The four lag operators perfectly align with $\omega(210) = 4$, generating an expansion zone that identically matches $\phi(210) = 48$. The 11 distinct residue classes that dictate schedule dissolution form a strictly graded meet-semilattice where the two maximal parity horns ($R8$ and $R9$) are topologically forbidden from joining, flawlessly validating the Parity Escape Law.

Finally, the advanced application of Markov Carry Chains [1] and the Triadic Role Invariant framework (WHAT, ON WHAT, HOW) proves that the preservation of this complex schedule geometry is internally absolute but strictly externally walled. The aggressive non-linear dynamics of the compression round, acting identically to an irreversible hydrodynamic vortex, systematically erase the parity birth-marks of the schedule, achieving a maximum-entropy readout. Thus, SHA-256 is fully understood not merely as an arbitrary data shredder, but as an elegant, continuous dynamic system operating on the inviolable constraints of addressed dependency geometry.

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